

Title

TEXT DATASET KNOWLEDGE → MODEL DECISIONS

A Complete Diagnostic Framework for Building Robust NLP Models

Top Workflow (same style)

1. MEASURE

- Analyze text properties
- Dataset statistics
- Language characteristics

→

2. DIAGNOSE

- Detect risks
- Data quality issues
- Bias & imbalance

→

3. PRESCRIBE

- Cleaning
- Tokenization
- Embeddings
- Model Selection

→

4. VALIDATE

- Evaluate improvements
 - NLP-specific metrics
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PART 1 — 22 KEY TEXT DATASET PROPERTIES

#	Category	Dataset Property	What to Measure	Why it Matters (Risk)	Common Fix
1	Dataset	Dataset Size	Documents, Sentences	Too little data → Overfitting	Data augmentation, Transfer Learning
2	Language	Language Distribution	English, Urdu, Mixed	Mixed languages confuse models	Language Detection
3	Language	Vocabulary Size	Unique words	Tiny vocabulary → Weak representation	Subword Tokenization
4	Text Length	Sentence Length	Average Tokens	Very long text exceeds context window	Chunking
5	Text Length	Document Length	Words per document	Memory issues	Sliding Window
6	Tokens	Token Distribution	Token frequency	Rare words poorly learned	BPE / WordPiece
7	Tokens	OOV Rate	Unknown Tokens	Embedding mismatch	Custom Tokenizer
8	Quality	Missing Text	Null documents	Training instability	Remove/Impute
9	Quality	Duplicate Documents	Exact/Near duplicates	Data leakage	Deduplicate
10	Quality	Label Noise	Wrong annotations	Incorrect learning	Manual Review
11	Class Balance	Class Distribution	Samples/Class	Minority ignored	Focal Loss
12	Labels	Multi-label Statistics	Labels/Document	Higher complexity	BCE Loss
13	Semantics	Class Overlap	Semantic similarity	Ambiguous labels	Better Annotation
14	Quality	Spelling Errors	Typos	Vocabulary explosion	Spell Correction
15	Quality	Special Characters	Emoji, HTML, URLs	Tokenizer problems	Cleaning Pipeline

#	Category	Dataset Property	What to Measure	Why it Matters (Risk)	Common Fix
16	Quality	Stopword Ratio	Stopword percentage	Weak signals	Selective Removal
17	Quality	Punctuation Distribution	Punctuation frequency	Noisy text	Normalization
18	Bias	Sensitive Attributes	Gender, Location etc.	Dataset bias	Balanced Sampling
19	Domain	Domain Shift	News vs Social Media	Poor generalization	Domain Adaptation
20	Temporal	Timestamp Drift	Old vs New data	Concept Drift	Incremental Training
21	Embedding	Embedding Coverage	Words in embedding	Unknown semantics	Fine-tune Embeddings
22	Readability	Reading Difficulty	Flesch, Grade Level	Complexity mismatch	Curriculum Learning

PART 2 — Evaluation Metrics

Classification

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- PR-AUC
- Confusion Matrix

Text Generation

- BLEU
- ROUGE
- METEOR
- BERTScore
- Perplexity

Question Answering

- Exact Match (EM)
- F1 Score

Named Entity Recognition

- Precision
- Recall
- Entity F1

Translation

- BLEU
 - chrF
 - COMET
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PART 3 — How Dataset Affects Model Decisions

Dataset Characteristic	Model Decision
Small Dataset	Transfer Learning
Large Dataset	Train from Scratch
Long Documents	Longformer / LED
Short Text	BERT / RoBERTa
Large Vocabulary	SentencePiece / BPE
High OOV Rate	WordPiece Tokenizer
Class Imbalance	Focal Loss
Multi-label	Sigmoid + BCE
Binary Classification	Single Output Sigmoid
Multi-class	Softmax
Noisy Labels	Label Smoothing
Multilingual Dataset	XLM-R, mBERT
Limited GPU	DistilBERT
Huge Dataset	DeBERTa, Llama, Mistral

PART 4 — Complete NLP Dataset Audit Pipeline

1 Dataset Audit

- Dataset Size
- Language
- Labels
- Balance
- Missing Values
- Duplicates

2 Text Statistics

- Vocabulary Size
 - Average Length
 - Token Distribution
 - OOV Rate
 - Stopwords
 - Readability
-

3 Label Statistics

- Class Counts
 - Multi-label Frequency
 - Class Overlap
-

4 Language Analysis

- Language Detection
 - Mixed Languages
 - Encoding
 - Character Distribution
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5 Data Quality

- Typos
 - HTML
 - URLs
 - Emojis
 - Noise
 - Duplicate Documents
-

6 Model Decisions

- Tokenizer
 - Embedding
 - Architecture
 - Context Length
 - Loss Function
 - Learning Rate
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7 Validation

- Accuracy
- F1
- BLEU
- ROUGE
- Perplexity
- Error Analysis

Key Takeaway

Understand your text before choosing your NLP model. Better dataset understanding leads to better tokenization, stronger representations, improved generalization, and higher real-world NLP performance.

I also recommend making this part of a **consistent infographic series**, all using the same professional layout and color palette. The sequence would be:

1.  Image Dataset Auditor (*completed*)
2.  Text Dataset Auditor
3.  Tabular Dataset Auditor
4.  Time-Series Dataset Auditor
5.  Audio Dataset Auditor
6.  Video Dataset Auditor
7.  Multimodal Dataset Auditor

This would form a comprehensive "**Universal Dataset Auditor**" portfolio where each dataset type has its own specialized diagnostic framework but shares a consistent visual design and methodology.